OPERATING SYSTEM

Concept Map

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COURSE NAME : OPERATING SYSTEM

COURSE CODE : CSA0406

Process Management, Memory Management, File System, and I/O System

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Process Management

Handles creation, scheduling and control of processes.

- Process creation & termination
- CPU scheduling
- Inter-process communication (IPC)
- Synchronization
- Deadlock handling

System Calls

- fork()
- exec()
- wait()
- exit()
- kill()

Process needs memory and memory supports processes

Memory Management

Allocates and manages memory for processes.

- Memory allocation
- Paging / Segmentation
- Virtual memory
- Swapping
- Memory protection

System Calls

- mmap()
- munmap()
- brk()
- malloc()
- free()

System Calls

Interface between user programs and OS components.

Processes access files and devices

Memory is used for buffers and I/O data

File System

Manages files and directories on storage devices.

- File management
- Directory structure
- Access control
- File permissions
- Disk space management

System Calls

- open()
- read()
- write()
- close()
- mkdir()

File system uses I/O for storage and access

I/O System

Controls input and output devices.

- Device drivers
- Device controllers
- I/O scheduling
- Buffering / Caching
- Spooling

System Calls

- read()
- write()
- ioctl()
- open()
- close()

All components interact with hardware through drivers

HARDWARE

CPU | Main Memory (RAM) | Storage Devices (HDD/SSD) | Input Devices (Keyboard, Mouse) | Output Devices (Display, Printer) | Network Devices (NIC)

Key Interdependencies

1. Process Management ↔ Memory Management
2. Process Management ↔ I/O System
3. Process Management ↔ File System
4. Memory Management ↔ File System
5. Memory Management ↔ I/O System
6. File System ↔ I/O System

Example Flow (System Call: read())

User Program → System Call read() → Process Management → File System → I/O System → Device Driver & Hardware → Data returned to Process

Legend

- System call interaction
- Component interaction
- Uses / supports
- Controls / manages